Lab 4 - Command Line Operation (August 24, 2026)

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Paste captured screenshots of the results for each question.

1. Accessing directory: http://linuxfoundation.s3-website-us-east-1.amazonaws.com/TIY/usingcd/index.html

2. Hard link
   a. Create a text file fileA.txt containing a few lines of text by running this command:
   `for i in {1..10}; do echo "Hello Linux $i"; done > fileA.txt`

```
parallels@ubuntu-gnu-linux-24-04-3:~$ for i in {1..10}; do echo "Hello Linux $i"
; done > fileA.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ cat fileA.txtx
cat: fileA.txtx: No such file or directory
parallels@ubuntu-gnu-linux-24-04-3:~$ cat fileA.txt
Hello Linux 1
Hello Linux 2
Hello Linux 3
Hello Linux 4
Hello Linux 5
Hello Linux 6
Hello Linux 7
Hello Linux 8
Hello Linux 9
Hello Linux 10
```

   b. Create a hard link fileB.txt to link to fileA.txt. List both files.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ ln fileA.txt fileB.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ ls -l fileA.txt fileB.txt
-rw-rw-r-- 2 parallels parallels 141 Aug 24 16:33 fileA.txt
-rw-rw-r-- 2 parallels parallels 141 Aug 24 16:33 fileB.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ diff fileA.txt fileB.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ rm fileA.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ cat fileB.txt
Hello Linux 1
Hello Linux 2
Hello Linux 3
Hello Linux 4
Hello Linux 5
Hello Linux 6
Hello Linux 7
Hello Linux 8
Hello Linux 9
Hello Linux 10
```

   c. Show contents of fileA.txt and fileB.txt

d. Delete fileA.txt and show content of fileB.txt

```
parallels@ubuntu-gnu-linux-24-04-3:~$ diff fileA.txt fileB.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ rm fileA.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ cat fileB.txt
Hello Linux 1
Hello Linux 2
Hello Linux 3
Hello Linux 4
Hello Linux 5
Hello Linux 6
Hello Linux 7
Hello Linux 8
Hello Linux 9
Hello Linux 10
```

e. Create a soft link as fileC.txt file to link to fileB.txt

```
parallels@ubuntu-gnu-linux-24-04-3:~$ pwd
/home/parallels
parallels@ubuntu-gnu-linux-24-04-3:~$ ln -s /home/parallels/fileA.txt /home/para
llels/fileB.txt
ln: failed to create symbolic link '/home/parallels/fileB.txt': File exists
parallels@ubuntu-gnu-linux-24-04-3:~$ ln -s /home/parallels/fileB.txt /home/para
llels/fileC.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ cat fileC.txt
Hello Linux 1
Hello Linux 2
Hello Linux 3
Hello Linux 4
Hello Linux 5
Hello Linux 6
Hello Linux 7
Hello Linux 8
Hello Linux 9
Hello Linux 10
```

f. Delete fileB.txt and show content of fileC.txt

```
parallels@ubuntu-gnu-linux-24-04-3:~$ pwd
/home/parallels
parallels@ubuntu-gnu-linux-24-04-3:~$ ln -s /home/parallels/fileA.txt /home/para
llels/fileB.txt
ln: failed to create symbolic link '/home/parallels/fileB.txt': File exists
parallels@ubuntu-gnu-linux-24-04-3:~$ ln -s /home/parallels/fileB.txt /home/para
llels/fileC.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ cat fileC.txt
Hello Linux 1
Hello Linux 2
Hello Linux 3
Hello Linux 4
Hello Linux 5
Hello Linux 6
Hello Linux 7
Hello Linux 8
Hello Linux 9
Hello Linux 10
parallels@ubuntu-gnu-linux-24-04-3:~$ rm fileB.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ cat fileC.txt
cat: fileC.txt: No such file or directory
parallels@ubuntu-gnu-linux-24-04-3:~$
```

3. Find the `init.d` directory, starting from /, and then create a symbolic link from within your home directory to this directory.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ whereis /init.d
init.d: /etc/init.d
parallels@ubuntu-gnu-linux-24-04-3:~$ pwd
/home/parallels
parallels@ubuntu-gnu-linux-24-04-3:~$ ln -s /etc/init.d /home/parallels/onit.d
parallels@ubuntu-gnu-linux-24-04-3:~$ ls onit.d
alsa-utils           console-setup.sh   kmod               spice-vdagent
anacron              cron               openvpn            sssd
apache2              cups               plymouth           sysstat
apache-htcacheclean  dbus               plymouth-log       ufw
apparmor             gdm3               procps             unattended-upgrades
apport               grub-common        rsync              uuidd
binfmt-support       kerneloops         saned              whoopsie
bluetooth            keyboard-setup.sh  speech-dispatcher  x11-common
parallels@ubuntu-gnu-linux-24-04-3:~$
```

4. Create sample.txt containing 200 lines of any text by using this command:
   `for i in {1..200}; do echo "test$i"; done > random_text.txt`
   a. Use cat command to view the file in each page

```
parallels@ubuntu-gnu-linux-24-04-3:~$ for i in {1..200}; do echo "test$i"; done
> random_text.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ cat random_text.txt
test1
test2
test3
test4
test5
test6
test7
test8
test9
test10
test11
test12
test13
test14
test15
test16
test17
test18
test19
test20
test21
```

b. Use cat command to view the file with the line numbers

```
parallels@ubuntu-gnu-linux-24-04-3:~$ cat -n random_text.txt
     1  test1
     2  test2
     3  test3
     4  test4
     5  test5
     6  test6
     7  test7
     8  test8
     9  test9
    10  test10
    11  test11
    12  test12
    13  test13
    14  test14
    15  test15
    16  test16
    17  test17
    18  test18
    19  test19
    20  test20
    21  test21
```

c. Use less command to view the file with the line numbers

```
     1 test1
     2 test2
     3 test3
     4 test4
     5 test5
     6 test6
     7 test7
     8 test8
     9 test9
    10 test10
    11 test11
    12 test12
    13 test13
    14 test14
    15 test15
    16 test16
    17 test17
    18 test18
    19 test19
    20 test20
    21 test21
    22 test22
    23 test23
random_text.txt
```

d. View content from the last line to the first line

```
test23
test22
test21
test20
test19
test18
test17
test16
test15
test14
test13
test12
test11
test10
test9
test8
test7
test6
test5
test4
test3
test2
test1
parallels@ubuntu-gnu-linux-24-04-3:~$
```

e. View first 5 lines of the file

```
parallels@ubuntu-gnu-linux-24-04-3:~$ head -5 random_text.txt
test1
test2
test3
test4
test5
```

f. View last 5 lines of the file

```
parallels@ubuntu-gnu-linux-24-04-3:~$ tail -5 random_text.txt
test196
test197
test198
test199
test200
```

5. Working with files and directories
   a. Create *file1* and *file2* (two empty files) in the home directory

```
parallels@ubuntu-gnu-linux-24-04-3:~$ touch file1 file2
parallels@ubuntu-gnu-linux-24-04-3:~$ ls
Desktop       fileC.txt       init.d       onit.d           retry.sh
Documents     file.html       key.txt      Pictures         snap
Downloads     fruits.sh       key.txt.pub  Public           software.csv
example.html  hello_world.py  Music        random-exit.py   Templates
file1         hosts.csv       myfile       random_text.txt  Videos
file2         index.html      new_dir      rename.sh        while.sh
```

   b. Check for the existence of *file1* and *file2*.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ touch file1 file2
parallels@ubuntu-gnu-linux-24-04-3:~$ ls
Desktop       fileC.txt       init.d       onit.d           retry.sh
Documents     file.html       key.txt      Pictures         snap
Downloads     fruits.sh       key.txt.pub  Public           software.csv
example.html  hello_world.py  Music        random-exit.py   Templates
file1         hosts.csv       myfile       random_text.txt  Videos
file2         index.html      new_dir      rename.sh        while.sh
```

   c. Rename *file1* to *new_file1*.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ ls -n new_file1
-rw-rw-r-- 2 1000 1000 0 Aug 24 16:45 new_file1
```

   d. Create a directory named *dir1*.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ mkdir dir1
parallels@ubuntu-gnu-linux-24-04-3:~$ ls -n dir1
total 0
```

   e. Move *new_file1* and *file2* files to *dir1* directory.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ mv new_file1 file2 dir1
parallels@ubuntu-gnu-linux-24-04-3:~$ ls dir1
file2  new_file1
```

f. Remove *new_file1* using `rm` without any options.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ cd dir1
parallels@ubuntu-gnu-linux-24-04-3:~/dir1$ rm new_file1
```

g. Remove *dir1* and its content.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ ls dir1
file2
parallels@ubuntu-gnu-linux-24-04-3:~$ rm dir1/file2
parallels@ubuntu-gnu-linux-24-04-3:~$ rmdir dir1
parallels@ubuntu-gnu-linux-24-04-3:~$ ls dir1
ls: cannot access 'dir1': No such file or directory
parallels@ubuntu-gnu-linux-24-04-3:~$
```

6. Create an empty file named `exercise.txt` and move this file to the `/tmp` directory, using a relative pathname from your home directory. Then, delete this file using an absolute pathname.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ touch exercise.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ mv exercise.text /tmp
mv: cannot stat 'exercise.text': No such file or directory
parallels@ubuntu-gnu-linux-24-04-3:~$ mv exercise.txt /tmp
parallels@ubuntu-gnu-linux-24-04-3:~$ rm /tmp/exercise.txt
parallels@ubuntu-gnu-linux-24-04-3:~$ cat /tmp/exercise.text
cat: /tmp/exercise.text: No such file or directory
```

7. Locating files: http://linuxfoundation.s3-website-us-east-1.amazonaws.com/TIY/usinglocate/index.html

8. Using wildcards with ls: http://linuxfoundation.s3-website-us-east-1.amazonaws.com/TIY/usinglswildcards/index.html